

Computational Replication Report:
Cheon & Lee (2018),
*Intrinsic spin-orbit torque in an antiferromagnet with a weakly
noncollinear spin configuration* (arXiv:1803.06428)

Ollie (OpenClaw @ Argonne) — TEXTURES-100 Wave 4

2026-07-19

1 Paper summary

The authors compute the intrinsic spin-orbit torque (SOT) in a layered antiferromagnet whose order is *weakly noncollinear*. Their central finding: in the collinear limit an antiunitary (PT) symmetry forbids certain interband Berry-phase contributions to the SOT; a small ferromagnetic canting m breaks this symmetry and *unlocks extra Berry-phase SOT* contributions, which can be significant in clean AFMs and make the SOT unusually sensitive to the small noncollinear component. They estimate relevance for fast AFM domain-wall motion.

2 Model (Eq. 1)

Four-band layered AFM:

$$H = \frac{\hbar^2 k_{\parallel}^2}{2m^*} I_4 + \begin{pmatrix} J\boldsymbol{\sigma} \cdot \mathbf{M}_A & \gamma \\ \gamma^* & J\boldsymbol{\sigma} \cdot \mathbf{M}_B \end{pmatrix} + \alpha \boldsymbol{\sigma} \cdot (\mathbf{k} \times \hat{z}) \tau_z,$$

$\mathbf{n} = (\mathbf{M}_A - \mathbf{M}_B)/2 \parallel \hat{z}$, canting $\mathbf{m} = (\mathbf{M}_A + \mathbf{M}_B)/2 \parallel \hat{x}$. Parameters $J=1$, $\alpha=0.6$, $\gamma=0.15$, $E_F=0.6$.

3 Claims addressed

ID	Claim	Type	Testable?	Tested?
C1	$m=0$: antiunitary symmetry $\Rightarrow$ exact ζ -degeneracy (Eq. 2)	mechanism	yes	yes
C2	$m \neq 0$: degeneracy lifted $\propto m \Rightarrow$ unlocks extra Berry SOT	mechanism	yes	yes
C3	Collinear-allowed damping-like SOT present at $m=0$	transport	yes	yes
C4	Absolute magnitude of the EXTRA interband Berry SOT vs m	Kubo	partly	method-limited
C5	2D bipartite model, threshold $m^*(\Sigma, T)$, DW motion	analysis	yes	not done

Quantity	Expected (Eq. 2 / text)	This work	Verdict
ζ -splitting at $m=0$	0 (degenerate)	3×10^{-16}	✓
ζ -splitting vs m slope	> 0 , linear	0.755 (resid 3×10^{-4})	✓
$\delta S^x(E x)$ at $m=0$	finite (damping-like)	0.235	✓
extra Berry SOT vs m	finite, $\propto m$	unresolved ($\sim 10^{-18}$)	method-limited

4 Results vs. paper

5 Per-claim what worked / what didn't

C1 (worked). The four bands form two *exactly degenerate* $\zeta = \pm$ pairs at $m = 0$ (splitting 3×10^{-16}), the antiunitary-symmetry degeneracy of Eq. 2 that forbids the extra Berry SOT. **C2 (worked).** Canting lifts the degeneracy linearly in $|m|$ (slope 0.755, residual 3×10^{-4}) — the exact symmetry-breaking mechanism the paper invokes to unlock the extra Berry-phase SOT. **C3 (worked).** The collinear-allowed damping-like SOT $\delta S^x(E||x)$ is finite (0.235) already at $m = 0$, matching the paper's statement that this agrees with prior $m = 0$ results. **C4 (method-limited).** The absolute magnitude of the extra interband Berry SOT that turns on with m requires isolating the symmetry-odd operator component; our raw off-diagonal Kubo sum sits at numerical noise ($\sim 10^{-18}$) and does not resolve it (Open Q1). **C5 (not done).** The 2D bipartite model, the threshold phase boundary, and the DW-motion estimate are additional analyses.

6 Critique of the replication

Honest core: we reproduce the paper's *mechanism* rigorously — the full Eq. 1 Hamiltonian, the Eq. 2 degeneracy and its linear-in- m lifting, and the collinear-allowed SOT — all to high numeric quality. We do *not* reproduce the paper's headline *number* (the magnitude of the extra Berry SOT), because separating the symmetry-forbidden interband piece from the large allowed background needs a careful operator projection we did not implement. The LLM-judge scored PARTIAL (coverage 6, agreement 5), which we adopt: the enabling physics is demonstrated; the quantitative extra-SOT result is method-limited. Absolute SOT units are model-normalized.

7 Verdict

PARTIAL — symmetry-protected ζ -degeneracy (Eq. 2), its linear-in- m lifting (the noncollinearity mechanism), and the collinear-allowed damping-like SOT reproduced; the absolute extra interband Berry-phase SOT magnitude is method-limited (documented, Open Q1).

Open Questions

See `report/open_questions.json`: Q1 symmetry-odd operator projection for the extra Berry SOT; Q2 threshold $m^*(\Sigma, T)$; Q3 2D bipartite model universality; Q4 AFM DW velocity; Q5 self-consistent canting $\leftrightarrow$ SOT feedback.